

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
Third Semester of B.Sc. (Physics) Examination November 2019
PD201 Mathematical Physics-II

Date: 06/11/2019, Wednesday Time: 10:00 a.m. to 10:30 a.m. Maximum Marks: 20

MCQ

Important Instructions:

1. Tick the correct answer and it should be written in question paper itself.
 2. You can use only non-programmable calculator. No other gadget is allowed in the examination hall.
 3. Each question carries one mark.
 4. All notations are standard.
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Q - I Choose the most appropriate answer from the given options for the followings:

1. The period of the function $f(x) = \cos\left(\frac{x}{3}\right) + \sin\left(\frac{x}{5}\right)$ is _____.
 - (a) 15π
 - (b) $\frac{\pi}{15}$
 - (c) $\frac{2\pi}{15}$
 - (d) 30π
2. $B\left(\frac{1}{2}, \frac{5}{2}\right)$ _____.
 - (a) $\frac{3\pi}{8}$
 - (b) $\frac{15\pi}{8}$
 - (c) $\frac{\pi}{8}$
 - (d) $\frac{3\sqrt{\pi}}{8}$
3. $\Gamma\left(\frac{-1}{2}\right) =$ _____.
 - (a) $\sqrt{\pi}$
 - (b) $-2\sqrt{\pi}$
 - (c) $-\sqrt{\pi}$
 - (d) $2\sqrt{\pi}$
4. The error function is _____ function.
 - (a) a odd
 - (b) neither even nor odd
 - (c) an even
 - (d) none of these
5. The Fourier series of an odd function contains _____ series.
 - (a) sine
 - (b) cosine
 - (c) both sine and cosine
 - (d) none of these
6. The differential equation $(2 + x^2)y'' + xy' + y = 0$ has singular points at _____.
 - (a) $x = \pm\sqrt{2}$
 - (b) $x = \pm 2$
 - (c) $x = \pm 4i$
 - (d) $x = \pm\sqrt{2}i$
7. The Hermite polynomial is a solution of _____.
 - (a) $x^2y'' + 2xy' + 4y = 0$
 - (b) $(1 - x^2)y'' + 2xy' + 4y = 0$
 - (c) $y'' - 2xy' + 4y = 0$
 - (d) $(1 - x^2)y'' - 2xy' + 4y = 0$
8. The differential equation $2(x - 1)y'' + xy' + y = 0$ has $x = 0$ as _____.
 - (a) a singular point
 - (b) an ordinary point
 - (c) a regular singular point
 - (d) none of these

9. The diffusion equation is given by _____.
 (a) $\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$ (c) $\frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial y^2}$
 (b) $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ (d) $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$
10. The function does not possess the Fourier series if f is _____.
 (a) continuous (c) discontinuous
 (b) non-periodic (d) periodic
11. The differential equation $25(1-x^2)y'' - 18xy' + 14y = 0$ is a _____ differential equation with order _____.
 (a) Legendre, $\frac{1}{5}$ (c) Legendre, $\frac{2}{5}$
 (b) Laguerre, $\frac{7}{25}$ (d) Laguerre, $\frac{2}{25}$
12. The generating function of the Bessel function is given by _____.
 (a) $e^{\frac{x}{2}(t-\frac{1}{t})}$ (c) $(1 - 2xt + t^2)^{\frac{1}{2}}$
 (b) $(1 - 2xt + t^2)^{-\frac{1}{2}}$ (d) $e^{-\frac{x}{2}(t-\frac{1}{t})}$
13. The function $\Gamma(z)$ is not defined _____ by Weierstrass definition.
 (a) for all $z \in \mathbb{Z}^+$ (c) for all $z = 0, -1, -2, -3, \dots$
 (b) for all $z = 0, 1, 2, 3, \dots$ (d) for all $z \in \mathbb{Q}$
14. As per the error propagation theorem, in addition and subtraction, a bound error of the result is given by the _____ of the error bounds for the terms.
 (a) subtraction (c) multiple
 (b) division (d) sum
15. For the linear curve fitting, when the data values are given large then _____ method is preferable.
 (a) step deviation (c) error propagation
 (b) direct (d) none of these

16. $\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} = \underline{\hspace{2cm}}$.
 (a) $\frac{\pi^2}{12}$ (b) $-\frac{\pi^2}{12}$ (c) $\frac{\pi^2}{6}$ (d) $\frac{\pi^2}{8}$
17. In the Fourier series expansion of the function $f(x) = \tan x$, $x \in (-\frac{\pi}{2}, \frac{\pi}{2})$ _____.
 (a) $b_n = 0$, $n = 1, 2, 3, \dots$ (c) $a_n = 0$, $n = 0, 1, 2, \dots$
 (b) $a_n \neq 0, b_n \neq 0$, $n = 1, 2, \dots$ (d) none of these
18. $\int_0^{\infty} e^{-\frac{x^2}{4}} dx = \underline{\hspace{2cm}}$.
 (a) $\frac{\sqrt{\pi}}{4}$ (b) $-\frac{\sqrt{\pi}}{4}$ (c) $\sqrt{\frac{\pi}{2}}$ (d) $\sqrt{\pi}$
19. The integral represented by $\int_x^{\infty} e^{-t^2} dt$ is _____.
 (a) $\frac{\sqrt{\pi}}{2} \operatorname{erfc}(x)$ (c) $\frac{\sqrt{\pi}}{2} \operatorname{erf}(x)$
 (b) $\frac{2}{\sqrt{\pi}} \operatorname{erfc}(x)$ (d) $\frac{2}{\sqrt{\pi}} \operatorname{erf}(x)$
20. The equation $10 \frac{\partial^2 z}{\partial x^2} + 10 \frac{\partial^2 z}{\partial y^2} - 20 \frac{\partial^2 z}{\partial x \partial y} - xy = 0$ is _____.
 (a) hyperbolic (c) elliptic
 (b) parabolic (d) none of these
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Third Semester of B.Sc. (Physics) Examination November 2019
PD201 Mathematical Physics-II

Date: 06/11/2019, Wednesday Time: 10:30 a.m. to 01:00 p.m. Maximum Marks: 50

Instructions:

1. **Section I and Section II must be written in separate answer books..**
2. You can use only non-programmable calculator. No other gadget is allowed in the examination hall.
3. Figures to the right indicate marks.
4. All notations are standard.

SECTION I

Q - II (A) Attempt any TWO of the following. [8]

- (1) Compute the Fourier series of a periodic function $f(x) = x^2$, $-L < x \leq L$.
 Use the series to compute the sum of the series $\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}$.

- (2) Find the straight line that fits the following data:
- | | | | | | |
|---|----|----|----|----|----|
| x | 1 | 2 | 3 | 4 | 5 |
| y | 14 | 27 | 40 | 55 | 68 |

- (3) Suppose $V = \frac{3a^2 + 5b^2}{c}$, where $a = (8.2 \pm 0.1)$ cm, $b = (6.5 \pm 0.1)$ cm, and $c = (5.1 \pm 0.1)$ cm. Compute the relative percentage error in V .

(B) Attempt any FOUR of the following. [12]

- (1) Obtain the relation between beta function and trigonometric functions.
- (2) Prove that $\Gamma(z+1) = z \Gamma(z)$, $z \in \mathbb{C}$ and hence evaluate $\Gamma\left(\frac{9}{2}\right)$.
- (3) Prove that $J_{\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}} \sin x$.
- (4) Solve the PDE $3p + 2q = 0$, where $p = \frac{\partial u}{\partial x}$ and $q = \frac{\partial u}{\partial y}$.
- (5) Find the half range Fourier cosine series of the function

$$f(x) = \begin{cases} 1; & 0 < x < \frac{\pi}{2} \\ 0; & \frac{\pi}{2} \leq x < \pi. \end{cases}$$

- (6) Find the approximate value of $f(x) = \sin 46^\circ \cos 58^\circ$.

SECTION II

Q - III (A) Attempt any THREE of the following.

[18]

- (1) (i) Evaluate $\int_3^7 \sqrt[4]{(x-3)(7-x)} dx$.
 (ii) Obtain the power series representation of the error function $erf(x)$ and use it to evaluate $erf(0.5)$.
- (2) Obtain the series solution of $xy'' + y' + xy = 0$.
- (3) Show that for $n \in \mathbb{N} \cup \{0\}$, (i) $P_n(1) = 1$ (ii) $P_n(-x) = (-1)^n P_n(x)$ and (iii) $P_n(-1) = (-1)^n$.
- (4) Compute the Fourier series of a periodic function

$$f(x) = \begin{cases} 0; & -\pi \leq x < 0 \\ \sin x; & 0 \leq x \leq \pi. \end{cases}$$

Hence find the sum of the series $\frac{1}{1 \cdot 3} + \frac{1}{3 \cdot 5} + \frac{1}{5 \cdot 7} + \frac{1}{7 \cdot 9} + \dots$.

- (5) A rod of 30cm long has its end A and B are kept at 20° and 80° respectively until the steady state condition prevail. The temperature at each end is suddenly reduced to 0° and kept so. Find the resulting temperature $u(x, t)$ from the end A.

(B) Attempt any TWO of the following.

[12]

- (1) (i) The power dissipated in a resistor is given by $P = \frac{E^2}{R}$. Using calculus find the approximate percentage change in P when E is increased by 3% and R is decreased by 2%.
 (ii) Obtain the recurrence relation $nP_n(x) = xP'_n(x) - P'_{n-1}(x)$, $n \in \mathbb{N} \cup \{0\}$.
- (2) Solve one dimensional heat equation $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$ with the boundary conditions $u(0, t) = u(\ell, t) = 0$, for all t and initial condition $u(x, 0) = f(x)$.
- (3) Find the Fourier series of the periodic square function

$$f(x) = \begin{cases} -k; & -\pi < x < 0 \\ k; & 0 < x < \pi. \end{cases}$$

. Hence deduce that $1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots = \frac{\pi}{4}$